

Problem Set 1

OE 754/854 — Ocean Waves & Tides

Assigned: end of Week 2 • Due: Friday of Week 3

This problem set exercises the fluid-mechanics machinery from Weeks 2 and 3 of the notes. Section references are to the lecture notes unless noted. The Week 2, 3, and 4 notebooks (`week02_fluid_forces.ipynb`, `week03_fluid_kinematics.ipynb`, `week04_bvp.ipynb`) demonstrate numerical counterparts of Problems 3–5: material-derivative tracking, Bernoulli along a converging channel, and the numerical Laplacian check.

Problem 1. Vector identities.

Let $\vec{a} = (a_x, a_y, a_z)$ and $\phi = \phi(x, y, z)$ be a differentiable scalar field. Prove, component by component, that

$$\vec{\nabla} \times (\vec{\nabla} \phi) = \vec{0} \quad \text{and} \quad \vec{\nabla} \cdot (\vec{\nabla} \times \vec{a}) = 0.$$

Irrotationality and incompressibility of potential flow rest on these two identities. We will use both within the first two weeks of wave theory.

Optional check. Verify the first identity on $\phi = x^2y + yz^2$. Verify the second on $\vec{a} = (yz, xz, xy)$.

Problem 2. Continuity equation.

Consider a small rectangular parcel of fluid with dimensions $\Delta x \times \Delta y \times \Delta z$ at position (x, y, z) , with constant density ρ . Derive the continuity equation

$$\vec{\nabla} \cdot \vec{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

by Taylor-expanding the mass flux across each of the six faces of the parcel. Balance the net flux against the rate of mass accumulation inside. Make the incompressibility assumption explicit and identify where it enters the derivation.

Hint. Follow the structure of §“Conservation of Mass” in the Week 3 notes.

Problem 3. Material derivative: three flow geometries.

For each of the following velocity fields, compute the material derivative $\frac{D}{Dt}u$ of the x -component at the indicated point. Identify which of the local acceleration $\partial u / \partial t$ and the advective acceleration $\vec{u} \cdot \vec{\nabla} u$ are nonzero.

- (a) **Time-varying tidal flow in a straight channel.** $\vec{u} = (U_0 \sin \omega t, 0, 0)$, with U_0, ω constants. Evaluate at $t = 0$.
- (b) **Steady flow through a converging nozzle of radius** $r(x) = r_0(1 - \alpha x)$. By continuity, the axial speed on the centerline is $u(x) = Q/[\pi r^2(x)]$ for constant volume flux Q , and $v = w = 0$. Evaluate $\frac{D}{Dt}u$ at $x = x_0$.
- (c) **Stagnation flow.** $\vec{u} = (kx, -ky, 0)$ with $k > 0$ constant. Evaluate at an arbitrary interior point $(x, y, 0)$.

Comment. Part (a) is pure local acceleration. Parts (b) and (c) are purely advective: each is a steady flow in which parcels still accelerate. All three can be measured on real flows. Think about what instrument would resolve each one.

Problem 4. Bernoulli's equation along a streamline.

Water flows steadily, inviscidly, and irrotationally in a horizontal converging nozzle. At the upstream section, the speed is $u_1 = 2$ m/s and the gauge pressure is $P_1 = 50$ kPa. The nozzle area contracts by a factor of 4 from upstream to the throat.

- (a) Find the speed u_2 at the throat.
- (b) Find the gauge pressure P_2 at the throat.
- (c) What would change in your analysis if the flow were unsteady? Write the extra term you would need to include and comment on when it is small.

Problem 5. Velocity potential for a point source.

The three-dimensional axisymmetric velocity potential for a "point source" of volumetric strength Q at the origin is

$$\phi(r) = +\frac{Q}{4\pi r}, \quad r = \sqrt{x^2 + y^2 + z^2}.$$

- (a) Compute the velocity field $\vec{u} = -\vec{\nabla}\phi$ and verify that it is purely radial: $\vec{u} = u_r(r)\hat{r}$. Give $u_r(r)$.
- (b) Verify $\nabla^2\phi = 0$ *everywhere except at the origin* by direct computation. (The source is at the singularity, $r = 0$.)
- (c) Compute the volume flux across a sphere of radius R centered on the origin and confirm that it equals Q , independent of R . (This is the physical meaning of the source strength Q .)

Problem 6 (Graduate students only). Vorticity equation.

Start from the Euler equations for an incompressible, inviscid fluid with body force $\vec{F} = -\rho\vec{\nabla}\Phi$ (conservative force, Φ = potential). Take the curl and show that the vorticity $\vec{\omega} = \vec{\nabla} \times \vec{u}$ satisfies

$$\frac{D}{Dt}\vec{\omega} = (\vec{\omega} \cdot \vec{\nabla})\vec{u}.$$

(This is the 3D inviscid vorticity equation. The right-hand side is the *vortex-stretching* term.) Then show that in a 2D flow ($\vec{u} = (u, v, 0)$, $\partial/\partial z = 0$), the vortex-stretching term vanishes and vorticity is materially conserved: $\frac{D}{Dt}\omega_z = 0$.

Submission. Neatly written or typeset, PDF. Show work for all derivations. Numerical answers should have appropriate units and three significant figures.